

Comparative assessment of human exposure to phthalate esters from house dust in China and the United States.

Because of volatilization and leaching from their application in consumer and personal care products, phthalate esters are ubiquitous contaminants in the indoor environment. In this study, we measured concentrations and profiles of 9 phthalate esters in indoor dust samples collected from six cities in China ($n = 75$). For comparison, we also analyzed samples collected from Albany, New York, USA ($n = 33$). The results indicated that concentrations, except for dicyclohexyl phthalate (DCHP) and bis(2-ethylhexyl) phthalate (DEHP), and profiles of phthalate esters varied significantly between the two countries. Concentrations of diethyl phthalate (DEP), di-n-hexyl phthalate (DNHP), and benzyl butyl phthalate (BzBP) were 5 to 10 times higher in dust samples collected from Albany than those from the Chinese cities. In contrast, concentrations of di-iso-butyl phthalate (DIBP) in dust samples from Albany were 5 times lower than those from the Chinese cities. We estimated the daily intake (DI) of phthalate esters through the routes of dust ingestion and dermal dust absorption. The extent of contribution of indoor dust to human exposures varied, depending on the type of phthalate esters. The contribution of dust to DEHP exposure was 2-5% and 10-58% of the estimated total DIs in China and the USA, respectively. On the basis of the estimates of total DIs of phthalates, extrapolated from urinary metabolite concentrations, the contributions of inhalation, dermal absorption, and dietary intake to total DIs were estimated. The results indicated that dietary intake is the main source of exposure to DEHP (especially in China), whereas dermal exposure was a major source for DEP. This is the first study to elucidate sources of human exposure to phthalates among the general population in China.